Operations Research 415
Part IV: Queuing Theory

Lecture 31: Finite Source Models

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Finite Source Models (p. 1099)

With the exception of the $M/M/1/GD/c/\infty$ model, all the systems considered thus far had arrival rates that are independent of the state of the system.

There are, however, two situations where the assumption of the state-independent arrival rates may be invalid:

1. If customers do not want to buck long lines, the arrival rate may be a decreasing function of the number of people present in the queuing system.
2. If arrivals to a system are drawn from a small population, the arrival rate may greatly depend on the state of the system. For example, if a bank has only 10 depositors, then at an instant when all depositors are in the bank, the arrival rate must be zero, while if fewer than 10 people are in bank, the arrival rate will be positive.

Models in which arrivals are drawn from a small population are called finite source models.

The Machine Repair Problem (pp. 1099–1100)

The system consists of K machines and R repair people.

At any time, a particular machine is in either good or bad condition.

The length of time that a machine remains in good condition follows an exponential distribution with rate λ.

Whenever a machine breaks down, the machine is sent to a repair centre consisting of R repair people (one person is required to repair a machine).

If $j \leq R$ machines are in bad condition, a machine that has just broken will immediately be assigned for repair, while if $j > R$ machines are broken, then $j - R$ machines will be waiting in a single line for a repair worker to become idle.

The time it takes to complete repairs on a broken machine is exponentially distributed with rate μ (so the mean repair time is $\frac{1}{\mu}$).

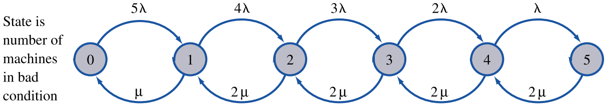
Once a machine is repaired, it returns to good condition and is again susceptible to breakdown.

The Machine Repair Problem (pp. 1099–1100)

Possible States in a Machine Repair Problem When $K = 5$ and $R = 2$

State	No. of Good Machines	Repair Queue	No. of Repair Workers Busy
0	$G\ G\ G\ G\ G$		0
1	$G\ G\ G\ G$		1
2	$G\ G\ G$		2
3	$G\ G$	B	2
4	G	$B\ B$	2
5		$B\ B\ B$	2

The *machine repair problem* is a $\boxed{\text{M/M/R/GD/K/K}}$ system



Taking

$$\lambda_j = \underbrace{\lambda + \lambda + \cdots + \lambda}_{K-j \text{ terms}} = (K-j)\lambda$$

and

$$\mu_j = \begin{cases} j\mu, & j = 0, 1, \ldots, R \\ R\mu, & j = R+1, R+2, \ldots, K \end{cases}$$

in the general birth-and-death process analysis (see Lecture 26), it can be shown that

$$\pi_j = \begin{cases} \binom{K}{j}\rho^j \pi_0, & j = 0, 1, \ldots, R \\ \frac{\binom{K}{j}\rho^j j! \pi_0}{R! R^{j-R}}, & j = R+1, R+2, \ldots, K \\ 0, & j = K+1, K+2, \ldots \end{cases}$$

where $\rho = \frac{\lambda}{\mu}$ and $\pi_0 + \pi_1 + \cdots + \pi_K = 1$.

The System $M/M/R/GD/K/K$ (pp. 1099–1100)

$$\begin{aligned}
\lambda &= \text{machine breakdown rate} \\
\mu &= \text{machine repair rate} \\
L &= \text{expected \# of broken machines} \\
L_q &= \text{expected \# of machines waiting for service} \\
W &= \text{average time a machine spends broken down} \\
W_q &= \text{average time a machine spends waiting for service}
\end{aligned}$$

$$\begin{aligned}
L &= \sum_{j=0}^{K} j\pi_j \quad \text{and} \quad W = \frac{L}{\overline{\lambda}} \\
L_q &= \sum_{j=R}^{K} (j-R)\pi_j \quad \text{and} \quad W_q = \frac{L_q}{\overline{\lambda}}
\end{aligned}$$

where

$$\overline{\lambda} = \sum_{j=0}^{K} \pi_j \lambda_j = \sum_{j=0}^{K} \lambda(K-j)\pi_j = \lambda(K-L)$$

is the average # of broken down machine arrivals per time unit.

Example 12 (pp. 1101–1102)

The Gotham Township Police Department has five patrol cars.

A patrol car breaks down and requires service once every 30 days.

The police department has two repair workers, each of whom takes an average of three days to repair a car.

Inter-breakdown times and repair times are exponentially distributed.

1. Determine the average number of police cars in good condition.
2. Find the average down time for a police car that needs repairs.
3. Find the fraction of the time a particular repair worker is idle.